

Samuel Shulman

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PROFESSIONAL SUMMARY

Software engineer passionate about full-stack development with hands-on experience across **React**, **Next.js**, **FastAPI**, and **AWS**. Currently building healthcare interoperability systems at Healthfirst while shipping side projects on nights and weekends. Self-motivated learner drawn to AI engineering and product-focused teams.

WORK EXPERIENCE

Technology Associate, Healthfirst

Sep 2025 – Present

- Engineered address-matching algorithm using **NLP**, **fuzzy matching**, and rule-based scoring; cut manual claim adjudication from **50%+** to under **10%** across **4M+** monthly claims.
- Built **Python** sync service between Workato and LeanIX, **automating** documentation updates across **500+ APIs** to eliminate stale architecture records.
- Architected Cost Estimator PoC using **AWS Step Functions** to **parallelize** provider lookups across **Lambdas**, resolving N+1 bottleneck.

PROJECTS

Real-Time Market Data Platform (shulman33.github.io/market-pipeline)

May 2026 – Present

- Architected a four-service microservices platform in **Python** and **Java** for real-time equity ingestion and order matching; containerized with **Docker**, deployed to **Amazon Web Services (AWS)**, and shipped through a GitHub Actions **CI/CD pipeline**.
- Engineered a limit-order-book matching engine in **Java** with price-time priority, applying **data structures and algorithms** (TreeMap, HashMap, doubly-linked FIFO queues) to keep order submission and cancellation fast; persisted to **PostgreSQL** via JDBC.
- Built an async **Python** WebSocket client against a **low-latency trade feed** with auto-reconnect and a stale-stream watchdog; flagged real-time price anomalies with a hand-rolled rolling z-score detector and wrote each tick atomically to **PostgreSQL**.
- Exposed the system via a **FastAPI REST** service and a Streamlit + Plotly dashboard; verified end-to-end through 46 **automated tests** with real **PostgreSQL integration tests**.

HelloMealio (hellomealio.com)

March 2026 – Present

- Engineered deterministic **greedy** + 1-swap algorithm to replace **Gemini LLM** meal planner; **250× faster (3s → 10ms)** and beat LLM quality in 13 of 15 head-to-head trials.
- Built AI pipeline with **AssemblyAI** and **Gemini** to extract structured recipe data from social media videos and shared links.
- Built single-call **Postgres RPC functions** to consolidate multi-statement transactions; reduced API write latency by **82–95%**, all writes under **300ms**.
- Integrated **Sentry** tracing across **React Native**, **FastAPI**, and **Postgres**; diagnosed and resolved production **latency bottlenecks** end-to-end.

Career Twin

Feb 2026 – Feb 2026

- Built **streaming AI chat agent** on **Next.js** using the **Vercel AI SDK** with Google Gemini, grounded in resume, LinkedIn, and personal context to answer in first person.
- Integrated Resend API as an autonomous tool call; agent forwards recruiter messages plus AI-generated summaries to my inbox for follow-up.
- Implemented **Langfuse** OpenTelemetry tracing across chat and summarization calls for end-to-end production **observability** of agent behavior.

ImIn

Jan 2023 – June 2023

- Engineered **Selenium**-based registration bot on **AWS Lambda**, **S3**, **DynamoDB**, and **EventBridge** that auto-enrolled students into classes within seconds of slots opening at YU.

- Designed **React** web app where students queued target classes; system handled login, form submission, and screenshot confirmation end-to-end.
- Achieved ~**50%** adoption across 2,300-student campus (**2,256 page views, 946 unique in 30 days**); university responded by mandating **MFA** on its registration system.

SKILLS

Languages: Python, React, Typescript, Javascript, HTML, CSS, SQL, Java

Technologies: Postgres, Next.js, FastAPI, Redis, AWS, Claude Code, Claude, Supabase, Vercel, Fly.io, Langchain

EDUCATION

Bachelor of Science, Computer Science, Yeshiva University

May 2025

- Relevant Coursework: Distributed Systems, Algorithms, Operating Systems, Modern Data Management, Compilers, Industrial Software Engineering, Data Structures, Networking, Cyber Security, Database Implementation, Programming languages

AWARDS AND ACCOLADES

- Yeshiva University Hackathon - Second Place

2023